



Best Practices for MVP300 Data Processing

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Abstract

This document provides a concise, actionable template for best practices in data processing for the MVP300 (Moving Vessel Profiler). The workflow includes steps for data acquisition, preprocessing, correction, calibration, and gridding, with a focus on ensuring data quality and reproducibility. A python module will be provided to implement the workflow and use it on MVP raw data in order to produce a standardised product, ready to be used by the scientific community. The goal is to ensure that data from the MVP300 are processed in a consistent, transparent, and reproducible manner, facilitating their use in scientific research and operational applications. **Keywords:** MVP300, Moving Vessel Profiler, oceanography, data processing, calibration, QA/QC, reproducibility

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1 Introduction

MVP300 is a state-of-the-art oceanographic instrument designed by AML Oceanographic for profiling water column properties while the vessel is underway. This device allows for high-resolution measurements of temperature, salinity, pressure, and sound velocity, providing critical data for oceanographic research and operational monitoring. However, the unique deployment context of the MVP300 generate different issues in data quality and processing compared to traditional CTD casts, necessitating tailored best practices to ensure data integrity and usability. The speed of the probe and its motions can introduce noise and biases that require specific preprocessing, correction, and calibration. This document outlines a comprehensive set of best practices for processing MVP300 data, covering all stages from acquisition to final product generation, with an emphasis on reproducibility and transparency.

2 MVP300 Overview

The MVP300 consist of an hydraulic winch system that deploys a profiling probe from the vessel's hull, allowing for continuous measurements as the ship moves. The probe is called "fish" and is equipped with a suite of sensors: a CTD, a sound velocity sensor, and optional sensors like fluorometer, oxygen, ph, SUNA sensors. Donwcast are done in free fall with a typical vertical speed of 2-3 m/s, reaching depths up to 3400m in stationary mode and 1250m at 6kn and 300m at 12kn. Here is a profile of the MVP300 vertical speed during upcast and downcast at 1000m.

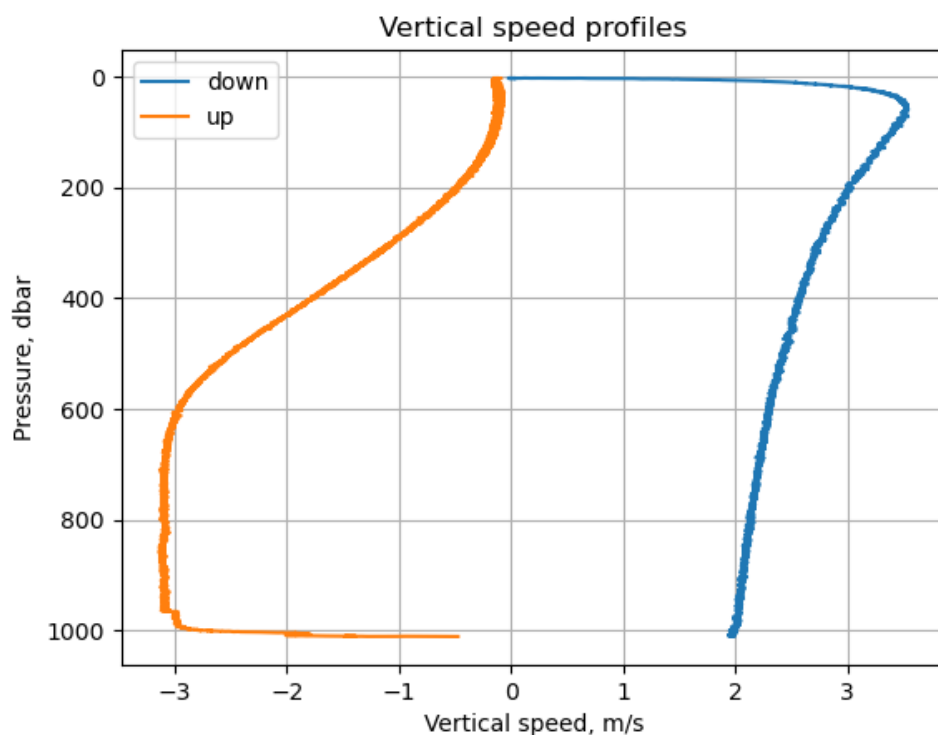


Figure 1: Vertical speed profile of MVP300 during upcast and downcast at 1000m depth.

For a standard deployment at 1000m with an horizontal speed of 4kn, the speed of the water flowing past the probe can reach 4 m/s, which can introduce significant noise and

biases in the measurements, particularly for conductivity and temperature. It is not easy to modelize and correct these effects, this is the purpose of the best practices described in this document, which are based on the experience of the team and the literature on similar devices like underway CTD (U-CTD) or gliders. By observation of [1](#), we can already stipulate that the upcasts present a better quality of data than downcasts: the higher vertical speed during downcast happen around the thermocline, which a critical layer where noising effect have a strong impact on the measurements. While during upcast, the vertical speed is more stable and lower on the thermocline, which can help to get better quality of data.

The MVP300 is equipped with an AML CT sensor (AML Oceanographic X2change™ Conductivity–Temperature (CT) sensor) [\[1\]](#), which is a relatively new technology that differs from traditional CTD sensors. With a double glass cross for temperature and conductivity. The glass cross technology is designed to minimize the thermal mass effect, and as the two glass crosses are very close, the same water particle is measured by both sensors. However there is still a temporal lag between the two measurements, which can introduce biases in the salinity computation. However, the response time of the conductivity cell is 25ms and the response time of the temperature cell is 100ms: these value that are quite high compared to other CTD sensors is surely the main reason of the temporal lag between conductivity and temperature.

A similar configuration of the MVP300 with AML CT sensor has been used by [\[5\]](#) however they did not apply any correction on the data.

3 Data Acquisition and preprocessing

Data Acquisition and postprocessing for the MVP300 are done by the AML software, which provides raw data files in ascii format, containing time series of sensor measurements along with metadata such as GPS position of each dive. Acquisition frequency is set to 20Hz for all sensors, using data duplication for sensors with lower native sampling rates to maintain synchronization. No filtering or correction is applied to the raw data, which is provided as-is for further processing. The raw data files are organized by dive, with one file with a downcast and a upcast, containing all measurements (temperature, conductivity, pressure) in a single file. There is one .log file with all metadata and one .raw file with all the measurements. In the .raw file, there is line benning by a Z with a suite of technical data not relevant for the scientific analysis. There is M lines with the measurements, each line corresponding to a scan of the sensors. There can also be some D lines with data from the SUNA sensor, as it need a full line to give the full spectrum of measurement.

4 Correction workflow

4.1 Manage and store raw data into up and down profiles

After getting the metadata from the .log and the data from the .raw files, the first step is to split the data into up and down profiles, using the pressure time series to identify the start and end of each profile. We chose to separate downcast and upcast at the first time the MVP reach the maximum of pressure for this profile, which corresponds to the end of the downcast and the start of the upcast.

Then we can store the data in a matrix form, with one matrix per variable and one line per profile (up for even index or down for odd index), and one column per measurement. In order to keep the matrix form, nan values are used to fill the empty cells for profiles with different number of measurements, which is the case for up and down profiles. These matrix can then be stored in a netcdf file, with one variable per matrix and one netcdf file per transect.

4.2 Use physical units

Once the profiles stored in matrix, we can convert electrical response of optional sensor to physical value. For CTD sensors (temperature, conductivity, pressure), the conversion is already done by the AML software, but for optional sensors (fluorometer, oxygen, ph, SUNA), the conversion is not done and we need to apply the conversion ourselves using the coefficients provided in calibration files.

Some sensors like the SUNA sensor for nitrate concentration, can't be converted to physical units using a simple linear conversion, but require a more complex method, as described in [?].

For oxygen, the value given by the sensor is in % saturation, and we need to convert it to $\mu\text{mol/L}$ using the solubility of oxygen in seawater, which depend on temperature, salinity and pressure. Thus, we will need to compute it after the correction of temperature, salinity and pressure.

A simple outlier detection is applied to the data, using threshold on the physical value of each variable, to remove any unrealistic values that may be due to sensor malfunction or data corruption. If we are in marine environment, we can use the following thresholds:

- Temperature: -2°C to 40°C
- Salinity: 0 to 40 PSU
- Pressure: 0 to 6000 dbar
- Oxygen: 0 to 400 $\mu\text{mol/kg}$
- Nitrate: 0 to 100 $\mu\text{mol/L}$
- pH: 6 to 9

4.3 Filter high frequencies

Lots of different element can introduce high frequencies noise in the measurements, such as the ship motion, the turbulence around the probe, the vibrations of the winch, the electrical noise etc... This noise can be reduced by applying a low-pass filter to the data, with a cutoff frequency finely tuned to preserve the physical information while removing the noise. For example, with a vertical speed of 2 m/s, if we want to preserve the information at a vertical scale of 2m (which is the case for the thermocline), we need to apply a low-pass filter with a cutoff frequency of 1Hz, which corresponds to a vertical scale of 2 m at a speed of 2 m/s. In our case we use a cutoff frequency of 0.1 Hz but with a low order filter (2) to preserve physical information.

4.4 Correct viscous heating

When a fluid exerts a friction thermistor of a temperature sensor, it can detect a temperature a bit higher than the one of the water around. As described in [?]:

$$dT = 0.80 \times 10^{-4} Pr^{0.5} v^2 \quad (1)$$

With Pr Prandti number and v vertical speed. This correction of temperature is quite low with an order of magnitude of 0.0001°C . As the resolution of the temperature sensor is 0.001°C , we can consider that this effect is negligible for the MVP300 and we will not apply any correction for it.

4.5 Correct temporal lag

One major correction to be made is the temporal lag between the measurement of conductivity and temperature. In fact, a slight shift between the two signal can have a great impact on salinity computation. In devices where water is pumped like CTD rosette or gliders, it is easy to determine the time a water particule take to go from the temperature cell to the conductivity cell. However in MVP300, water is not pumped and the sensor is different from the other MVP as well. The CT sensor is an AML product [1] relatively new at this time and not already widely used. The technology of this sensor is different from other sensors: The water flow through a glass cross which contains the electrodes. One glass cross for temperature cell and another for conductivity cell. So the same particule is not measured by the two sensors, which should not be a problem, as the two glass cross are very close, but we can't use the different method like [?] which stipulate that the water is the same in both cells. Moreover, as we saw earlier the response time for C and T are different and quite high, which can introduce another temporal lag between the two measurements. As a matter of fact, temporal lag is more challenging with this device, as we will determine it blindly.

The method used here for determine the temporal lag is the following:

1. Filter with a bandwidth filter temperature and conductivity to get the high frequencies information of temperature variation but without the noise
2. normalize both signals (temperature and conductivity)
3. compute cross-correlation between the signals for different time shift
4. The cross-correlation is maximised when the shift equal the temporal lag

For MVP 300 we find a temporal lag around 3 ou 5 scan, at 20 Hz, it correspond to a temporal lag of 0.15 to 0.25 second, which is quite high compared to other devices, but it is not surprising given the response time of the sensors.

Other methods exist, for example, as used by [2] which use max correlation between differenced temperature and conductivity with a dependance of the vertical speed. Or [3] minimizing the standard deviation of the salinity gradient. Our method may be enhanced, but it is simple and robust, and give good results for the MVP300.

4.6 Resample

In order to have proper data erasing the heaving of the device, it is necessary to resample the data on a regular pressure grid. The step should be finely tuned to preserve the physical information while avoiding to oversample the signal and introduce noise. For example at 20 Hz for raw data, with a max vertical speed of 3.5 m/s, we have a vertical resolution of 0.175 m. So we should not resample with a step smaller than 0.175 m, otherwise we will oversample the signal and introduce noise. A new pressure grid of 0.2 m should be very good to get all the smaller scale of physical effect.

4.7 Correct thermal mass effect

For CTD data it is common to apply a correction for the thermal mass effect. This effect is due to a local heating of the conductivity cell which can introduce a bias in the conductivity measurement. In strong gradient of temperature, this effect can be significant as the conductivity cell keep the memory of the previous temperature measurement, and so, the cell will heat the the sampled water which will be at as different temperature than the surrounding water and this introduce a bias in the conductivity measurement and therefore in the salinity computation.

One of the first to correct this effect was proposed by [4] and is based on a correction of conductivity using the difference of temperature between the current measurement and the previous one, with a coefficient which depend on the temperature sensibility of the sensor γ . The formula also use the previous conductivity measurement and two parameters a and b determined by optimization based on the minimization of the difference between up and down T-S curves.

$$C_i = -bC_{i-1} + a\gamma(T_i - T_{i-1}) \quad (2)$$

(the quation is implified here as a and bb are more complex than this, but the principle is the same)

Then [2] proposed a method correcting the thermal mass effect by modifying the temperature measurement instead of the conductivity measurement, with a similar formula but with a different coefficient.

$$T_i = -bT_{i-1} + a(T_i - T_{i-1}) \quad (3)$$

This method present the advantge to avoid using the sensibility coefficient of the sensor, which can be difficult to determine. a and b are determined by the same optimization as for the previous method, based on the minimization of the difference between up and down T-S curves. They are linked to the vertical speed of the probe, the sampling frequency and b depend on a .

However, these methods are based on the assumption that the thermal mass effect is a linear effect, which may not be the case, especially for the MVP300 which have a different technology of sensor and a higher vertical speed than other devices. In the case if the MVP300 and AML CT sensor, these method have been tested but they don't give good results, and we didn't find a way to modelise andcorrect the thermal mass effect for this device. It is complicated to determine the impact of the thermal mass effect on the measurements, as it can be confounded with other effects such as the lag between conductivity and temperature. Moreover the previous correction done on the temporal lag can already correct the thermal mass effect, as the lag is determined by maximizing the correlation between temperature and conductivity, which can be affected by the thermal

mass effect. So it remains an open question for now. Is the correction of the thermal mass effect necessary for the MVP300 ? If yes, how to modelise it and correct it ?

4.8 Median filtering

After all the previous corrections, it can be useful to apply a median filter to the data, with a window finely tuned to preserve the physical information while removing the remaining noise. For example, with a resampling at 5 sample / meter we will use a median filter with a window of 5 sample, which correspond to a vertical scale of 1 m. we only use this median filter for salinity profile as it is the most affected by noise.

4.9 Results

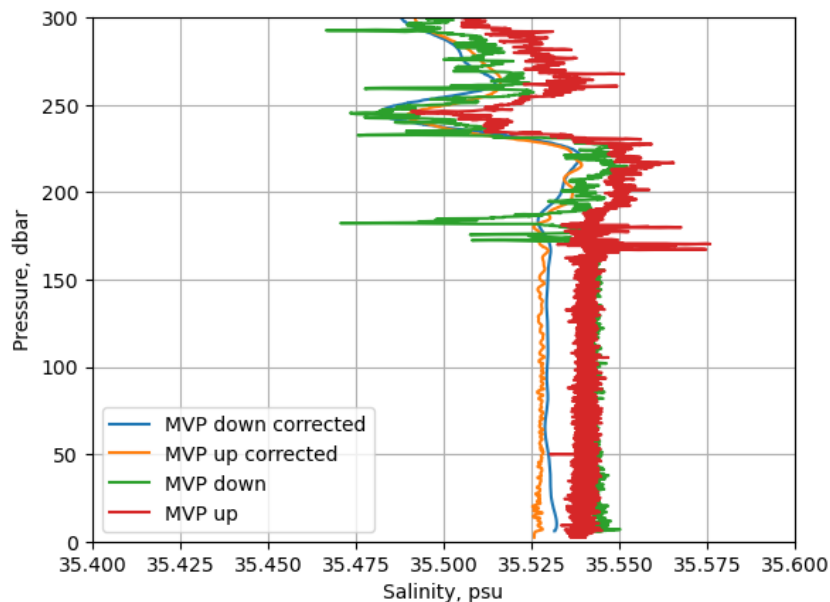


Figure 2: Example of salinity profile before and after correction.

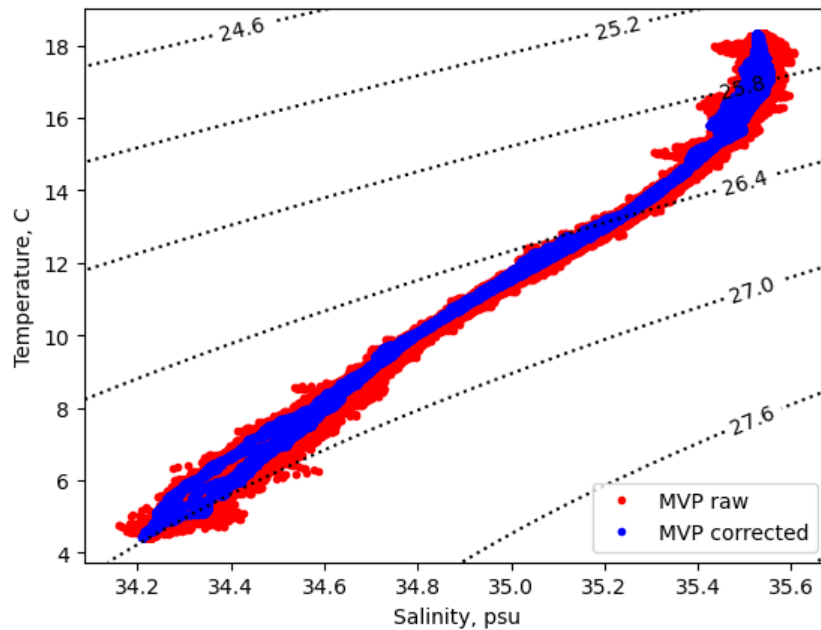


Figure 3: Example of T-S diagram before and after correction.

5 correction of biogeochemical sensors

5.1 Correction of optial sensors like SUNA sensor for nitrate concentration

We observed on all optical sensors, a strong difference between up and down profiles, which is not physical and can be due to the fact that the optical sensors are more affected by the turbulence around the probe, which is different for up and down profiles. We developed a method to correct this effect on the SUNA sensor for nitrate concentration. By plotting the difference between up and down profiles, we observed that the difference is not constant with depth, but it is a function of the depth. So we can fit a polynomial function to the difference between up and down profiles, and use this function to correct the down profile, by adding the fitted function to the down profile. This method could be applied to other optical sensors as well, such as fluorometer or ph sensors, as they are also affected by the turbulence around the probe.

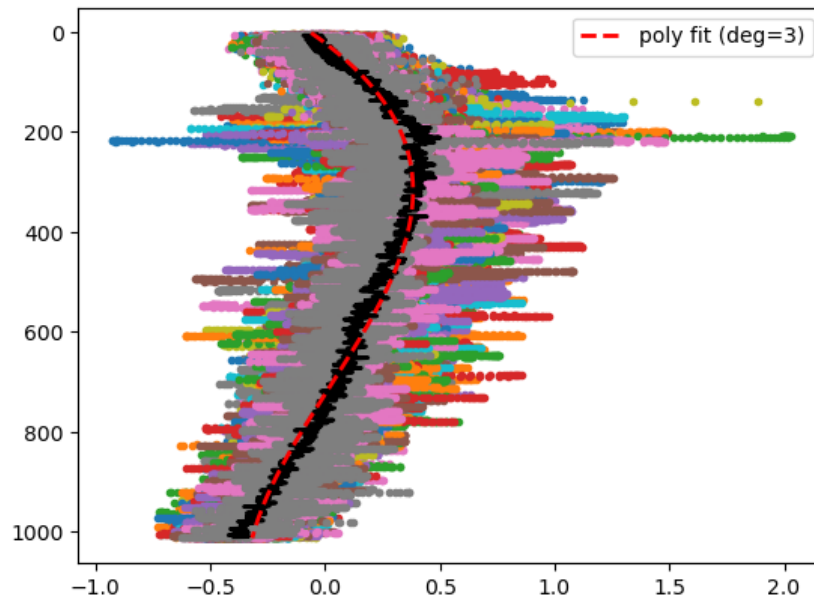


Figure 4: Example of nitrate concentration profile of error between downcast and upcast measurements as a function of pressure.

5.2 Correction of oxygen sensor RINKO3

The Rinko3 oxygen sensor is a fast response sensor, but there is a pressure effect on the measurement, which is not physical and due to the fact that the optode warps with pressure. Fortunately, the pressure effect is a linear function of pressure, and we can correct it by fitting a linear function to the difference between the Rinko3 measurement and the oxygen concentration computed from another sensor (e.g., CTD) using the same data, and use this function to correct the Rinko3 measurement.

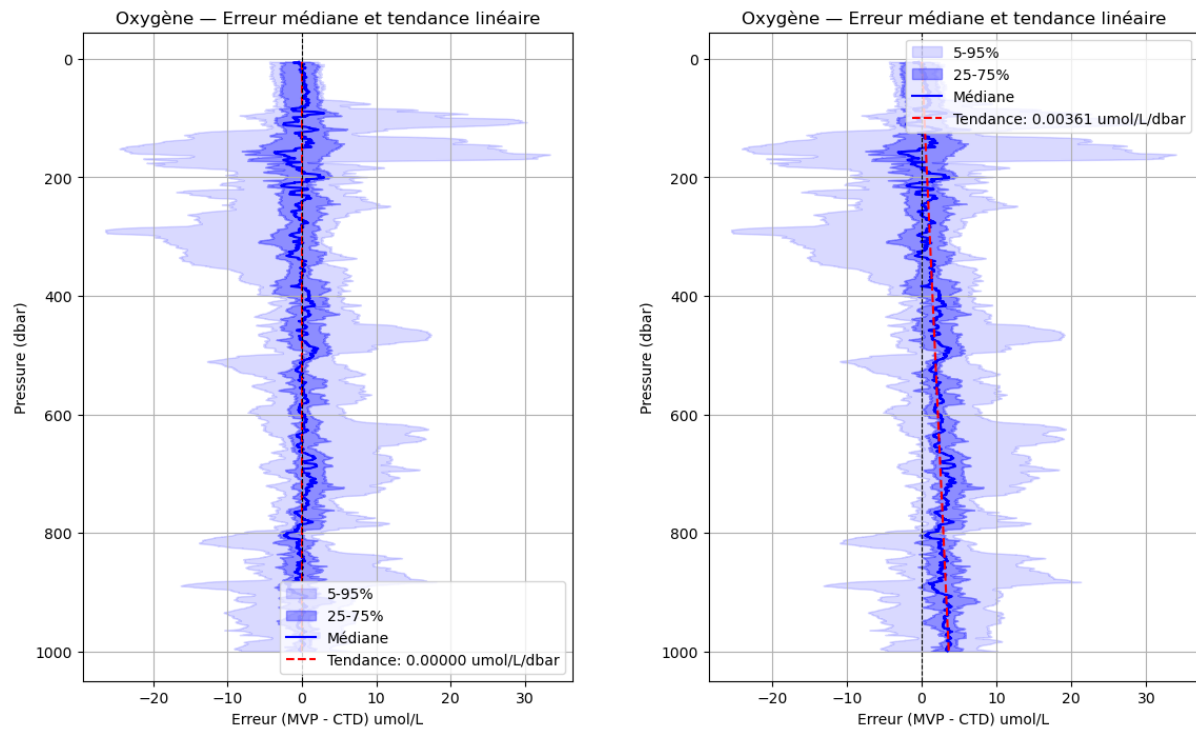


Figure 5: Visualization of the error between CTD and MVP measurements for oxygen, before and after slope correction

6 Calibration with nearby CTD

Nearby CTD casts can be used to calibrate the MVP300 data, in the same way niskin bottles are used to calibrate CTD data. Looking at the difference between the MVP300 data and the nearby CTD data, we can determine a bias for each variable (temperature, salinity, oxygen, nitrate, ph) and apply this bias to the MVP300 data.

Two different approaches can be used for the calibration: correct a MVP profile thanks to the nearest CTD profile, or correct all MVP profiles with a single bias determined by the average of the difference between all MVP profiles and the nearby CTD profiles. The second approach is more robust, as it is less sensitive to the noise of a single profile, but it can be less accurate if the bias is not constant with time or space. In order to ensure the quality of the calibration, it is important to have a good number of coupled MVP and CTD profiles, and to check the consistency of the bias with time and space. For this, we can look at the distribution of the difference between MVP and CTD profiles, and check if it is centered around a constant value or if it varies with time or space.

For example we can look at the data from the MVP300 deployed during the WHIRLS cruise. 870 MVP profiles were collected and 105 CTD profiles were collected. We have 15 coupled MVP and CTD profiles, which is enough data to be consistent and determine a robust bias for each variable.

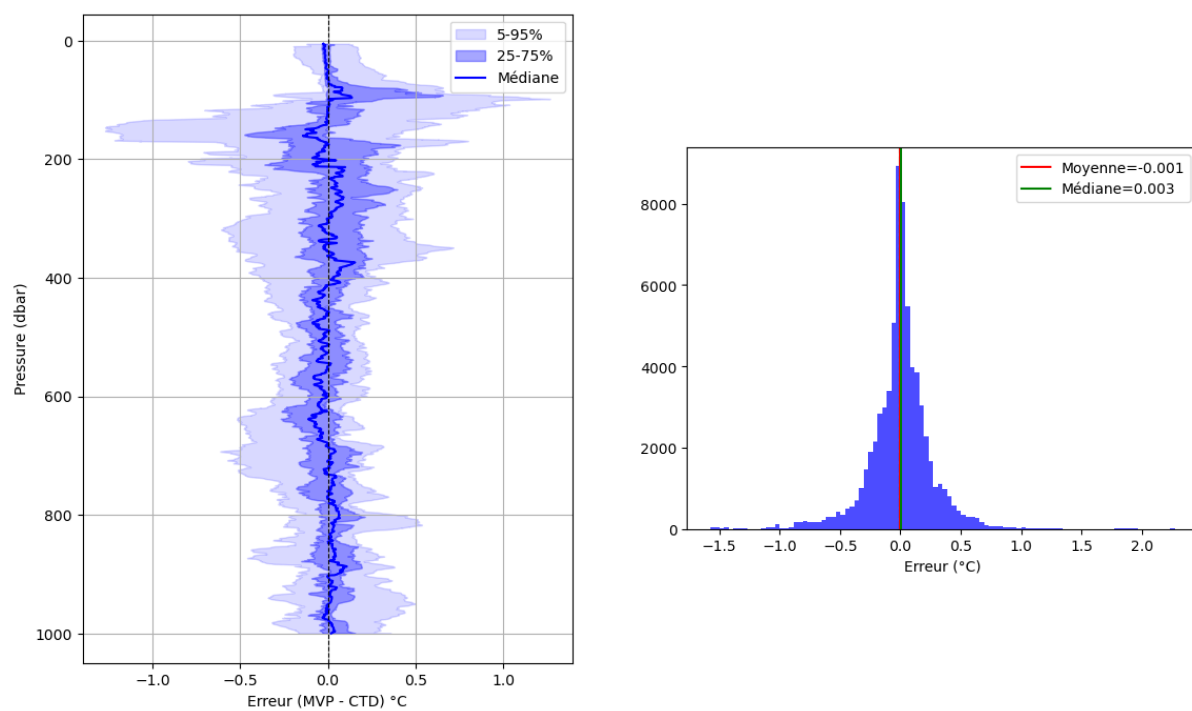


Figure 6: Visualization of the error between CTD and MVP measurements for temperature

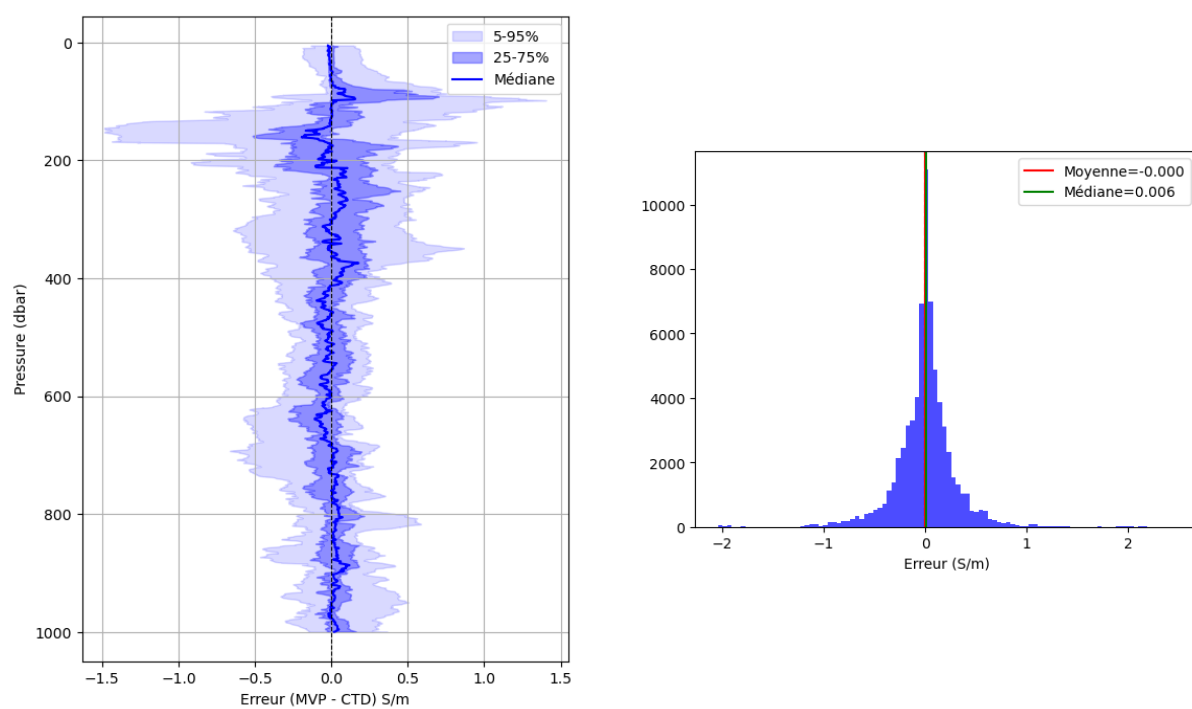


Figure 7: Visualization of the error between CTD and MVP measurements for conductivity

We can also observe that the error of salinity has a bias that disappears when we apply the offset to conductivity and temperature.

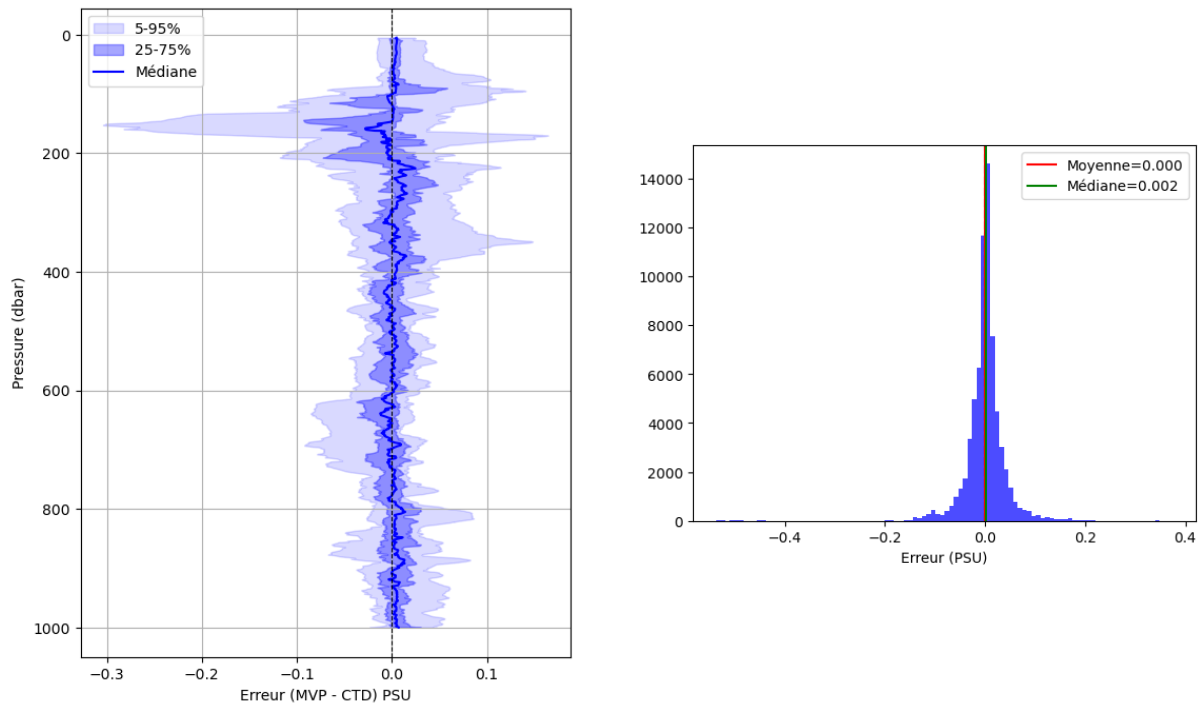


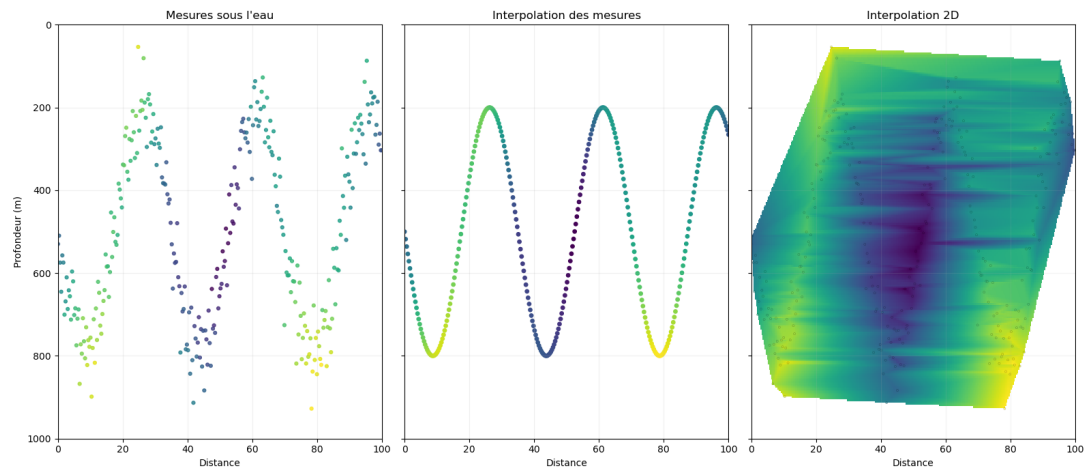
Figure 8: Visualization of the error between CTD and MVP measurements for salinity

7 2D interpolation and gridding

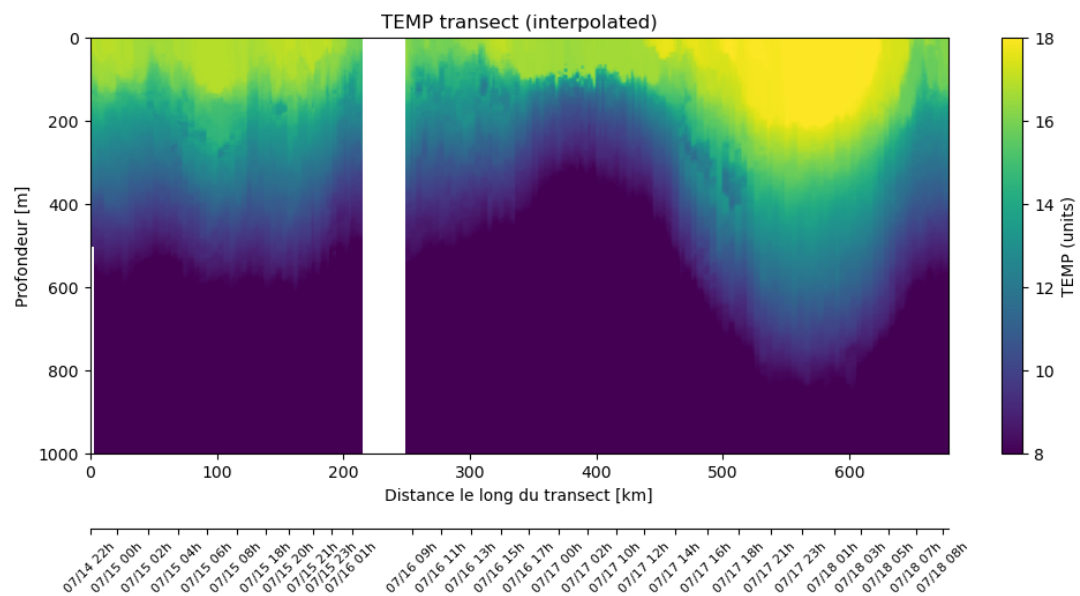
Once the MVP data are processed and calibrated, we can interpolate them on a regular grid in space and time, to produce a 2D product that can be used for further analysis and visualization. For this we choose as X-axis the distance along the transect, and as Y-axis the depth.

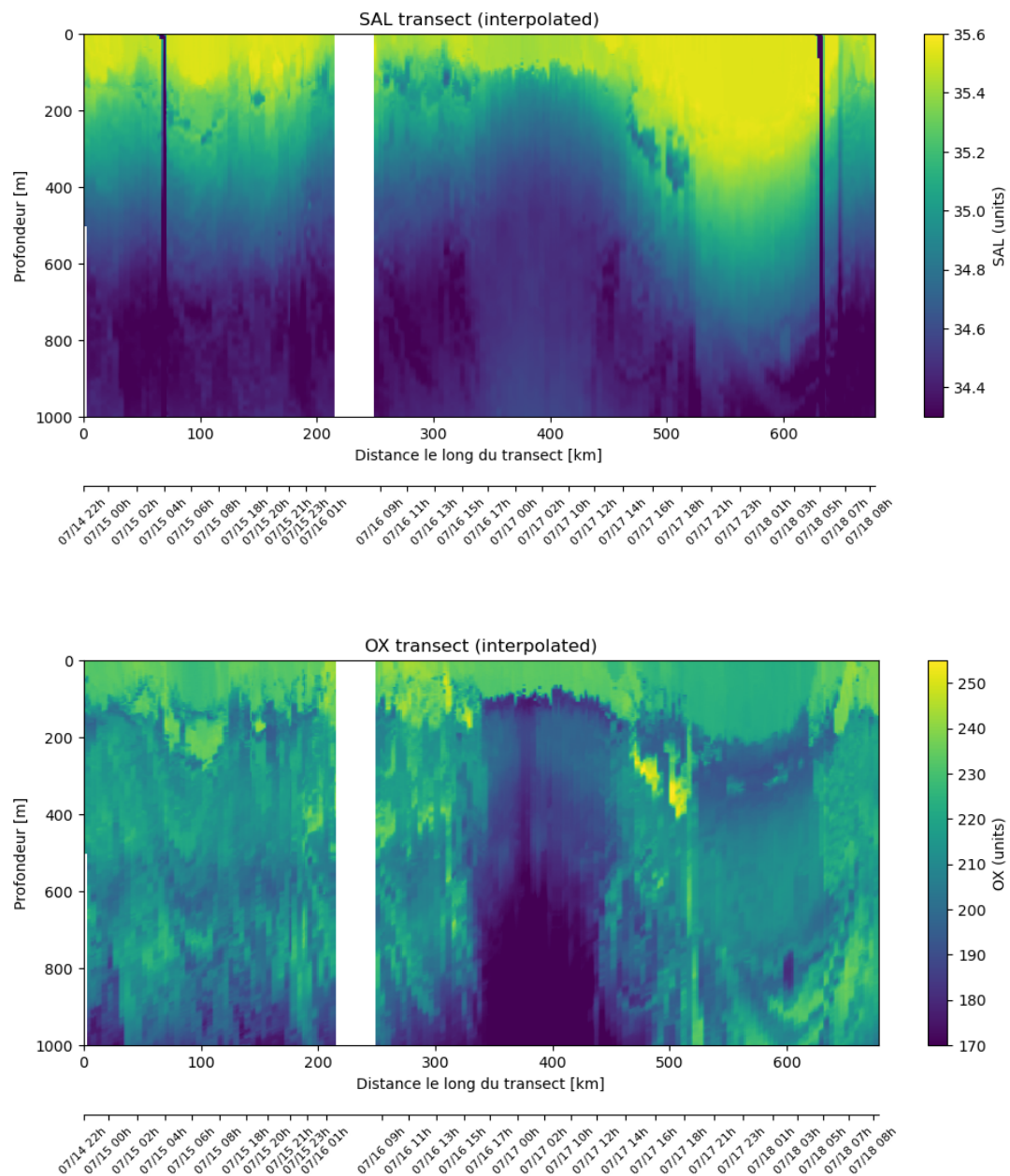
- We do not consider the MVP profile as vertical, as there is around 3.5 km of horizontal distance between the beginning of the downcast and the end of the upcast. We therefore obtain the GPS position of each MVP point using the ship GPS position.
- We then compute the distance between the first and the last point of an MVP profile, and use this distance to compute the distance between each MVP point and the first point of the section.
- MVP measurements are already interpolated on a regular pressure grid (see the section on MVP correction). We interpolate the data to obtain a 2D grid for each variable, with distance and pressure as coordinates. For this, we use linear interpolation at each pressure level.

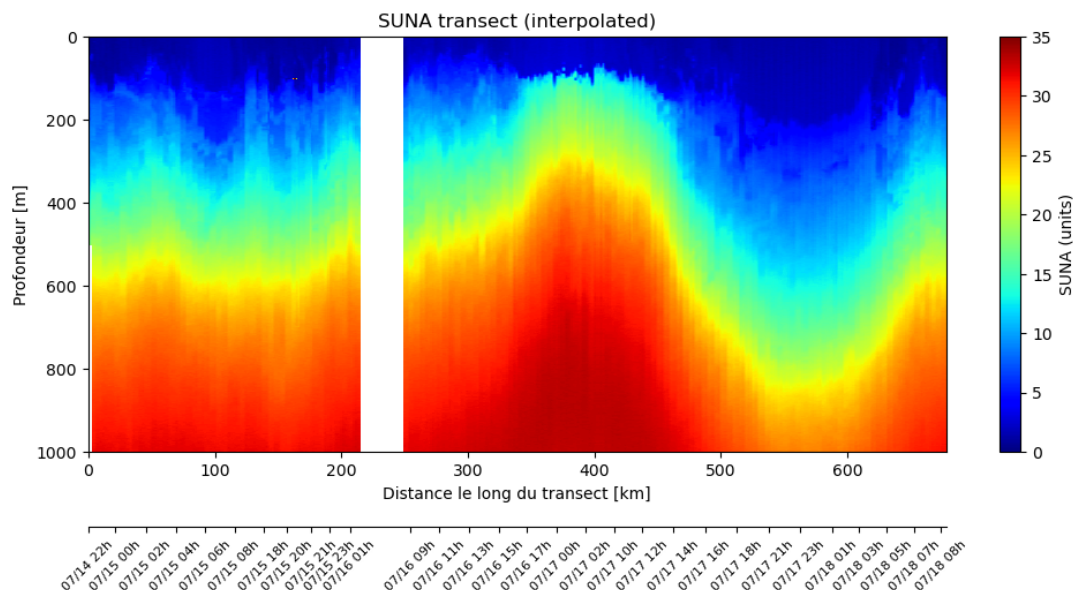
The following figures illustrate the process of the interpolation:



Here are some examples of the visualizations we can obtain with the corrected and calibrated MVP data: (transect 9,10,11 and 12 concatenated)







8 PyMVP: Python package for MVP300 data processing

In order to facilitate the application of these best practices, an open source Python package has been developed that implements the processing workflow described in this document. The package provides functions for each step of the processing, from raw data parsing to final product generation, with clear documentation and examples. The package is designed to be easily understood, used and modified by the scientific community, and it is available on GitHub <https://github.com/MxWmr/PyMVP> and downloadable as a Python package via pip: `pip install PyMVP`.

9 Acknowledgements

References

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